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EOSC-ES-Prep: preparing the Spanish EOSC Node

Deliverable 3 – Draft presentation for the showcase event (revised version, 28 September 2026)

Funded by the European Union

Grant agreement ID: 101188045

Project title: EOSC-ES-Prep

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Partners: BSC-CNS · CESGA · CIEMAT · CSIC · FIMABIS · IBiS-FISEVI · IBiS-Hospital Universitario Virgen Macarena · IBiS-Hospital Universitario Virgen del Rocío · IdISBa · Instituto de Investigación Sanitaria Biogipuzkoa · IMIB/FFIS · IRYCIS · RedIRIS (Red.es) · SCAYLE · Universidad de Alicante · Universidad de Córdoba · Universidad de Jaén · Universidad de Murcia · Universidad Politécnica de Madrid · Universitat Politècnica de València · Universidad del País Vasco / Euskal Herriko Unibertsitatea

Preparatory grant EOSCGRAVITY-PREP-2026-010 · 6 May – 6 November 2026 · 22 institutions

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Where we are, and what November will show

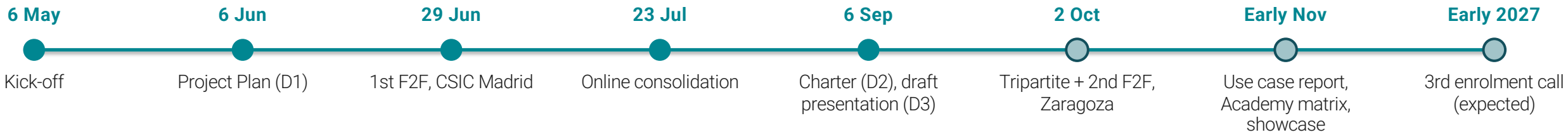
Status at month 5 of the preparatory grant (6 May – 6 November 2026)

Where we are (September 2026)

- **Project Charter (D2) submitted** on 6 September and **approved with recommendations**
- **Four use cases** committed for the Node build-up phase
  - Health, ocean, astrophysics, social sciences and humanities
- Governance model **agreed among the 22 institutions**
  - Existing entity as signatory plus an open collaboration agreement)
- **National tripartite meeting on 2 October 2026** in Zaragoza with the EOSC Association, the European Commission, the Spanish EOSC Steering Board delegate and COS-Gob.

What the November showcase will present

- **EOSC-ES Charter** as the implementation plan of the Spanish Node
  - Updated with agreed: governance, representing entity, role holders and core-service providers
- **Exemplar** use case.
- **Training activities of the consortium**, their readiness for the EOSC Academy and the skills the Node still needs.
- **Engagement results** measured against the **KPIs of the Project** Plan (D1), and the roadmap to the third-wave enrolment application.



Teal: completed · Orange: upcoming

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Objectives and outputs achieved

The three objectives of the Project Plan (D1) and the evidence for each

1. Map the Spanish ecosystem against Federation requirements

- **28 resources (services, repositories)**
  - 12 TRL7 or more
  - Collected through a form
- Mapping **between Federating Capabilities** and operating national system
  - RedIRIS SIR (FC-1); RECOLECTA, datos.gob.es and PLATICA (FC-2); Zammad at UPV/IACS (FC-3); ARGO (FC-4); FitSM (FC-5).

D1 KPIs 1 and 3 met: 22 institutions mapped (target 10); pathways documented for the five capabilities.

2. Define an inclusive governance

Agreed governance model

- Informed by the German (NFDI), Czech (EOSC-CZ) and European Digital Twin of the Ocean Nodes
- An existing entity **signs the Federation MoU on behalf of an collaboration agreement** with a general assembly
- **Draft collaboration agreement** (Joint Research Unit EOSC-ES). Representing entity is decided with the national authorities, **starting at the 2 October tripartite**.

3. Portfolio, use cases and Federation interactions

Project Charter (D2)

- Value proposition, 12 resources, 4 use cases, compliance with FC-1 to FC-5, 12 risks with mitigation, 7 build-up deliverables and 16 milestones over 24 months.

Four use cases

- All with counterpart EOSC Nodes and linked to Common European Data Spaces (health, Green Deal marine, language, cultural heritage).

D1 KPI 5 exceeded: target was 2 intra-node and 1 inter-node use case.

Deliverables to date: D1 Project Plan (June 2026) · D2 Project Charter (September 2026) · D3 draft showcase presentation (this document). Due early November: use case report, EOSC Academy matrix and showcase contribution.

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In-kind contribution and funding synergies

How the Node is built **without dedicated funds**, and where **build-up funding** can come from

In kind, today

Every resource proposed to the Node stays operated and **financed by its owner**, under budgets that already exist:

- RedIRIS / Red.es for identity federation (FC-1)
- FECYT / RECOLECTA for catalogue harvesting (FC-2)
- RES centres BSC-CNS, CESGA and SCAYLE, and PIC/CIEMAT, for compute and storage
- Spanish nodes of ELIXIR, BBMRI, ECRIN, EATRIS, CLARIN and DARIAH
- Regional platforms with their own mandate, such as BIGAN

Synergies with running projects

Use cases and capabilities reuse results of projects in which consortium partners already work:

- **EOSC United** — EHDS pilot (IACS), **EOSC ENTRUST** (BSC-CNS), **EOSC-SIESTA** (CSIC), **EUCAIM** (UPV) → ES-UC1
- **AI4EOSC** (CSIC) → AI services for ES-UC1 and ES-UC2
- **Blue-Cloud** 2026, already engaged with NEREIDAS (UMU) → ES-UC2
- **ESCAPE Data Lake** (PIC) → ES-UC3
- **ParlaMint** (CLARIN) through CLARIAH-ES → ES-UC4
- **EOSC Beyond** (monitoring) → FC-4
- **FAIR2Adapt** (UPM) → ES-RES03

Funding for build-up and operation

EU co-funding: HORIZON-INFRA-2027-01-EOSC-01 (COFUND, “Expanding and deepening the EOSC Federation”), for which the applicant is a national funding agency.

National policy: ENCA 2023–2027 and PEICTI 2024–2027 prioritise integration into EOSC; the Node is their implementation instrument.

Consortium: cost-sharing and membership rules in the collaboration agreement; service fees where resources already charge them.

Long term: inclusion of EOSC-ES in the national science, technology and innovation plans for 2028–2034.

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Use cases: overview

Four use cases the Node commits to develop during its build-up phase

Use case	Domain · Common European Data Space	Access regime	Counterpart EOSC Nodes	Lead	End-to-end demonstration
ES-UC1 Health data	Health, animal and environmental health · European Health Data Space	Governed (statutory secondary use)	Life Sciences Connect; BBMRI-ERIC; EOSC EU Node	IACS	M18 (governed-access rehearsal M15)
ES-UC2 Oceans	Marine sciences · Green Deal data space, marine component	Open and registered users	European Digital Twin of the Ocean; Blue-Cloud 2026 / CNR	CSIC, with UMU	M18
ES-UC3 Astrophysics	Astrophysics · none sectoral (EOSC as transverse data space)	Controlled (collaboration policy)	CERN Node, through the ESCAPE Data Lake	PIC / CIEMAT	M12 (workspace in operation M9)
ES-UC4 Parliamentary corpora	Social sciences and humanities · language and cultural heritage data spaces	Open	EOSC Node Netherlands (SURF); EOSC Node Slovenia; CLARIN and DARIAH ERICs	HiTZ (UPV/EHU) for CLARIAH-ES	M12

Common pattern

Discover in the Node catalogue (FC-2) → **Authenticate** (FC-1, SIR / eduGAIN) → **Analyse where the data reside** → **Publish** citable FAIR research products

Will they be developed? Implementation plans **for all four at build-up month 3** (ES-D5.0, ES-MS3); ES-UC3 and ES-UC4 demonstrated end to end at month 12 and ES-UC1 and ES-UC2 at month 18, each with a public report (ES-D5a/b). Three of them link EOSC

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ES-UC1 · EHDS: health data, One Health scope

Secondary use of health data that stay with their holders, across regions and domains

The challenge

Human, animal and **environmental health data sit under different regimes and custodians**. The EHDS Regulation creates a **statutory framework for their secondary use**, and its **catalogue** already spans environmental and behavioural determinants of health and pathogens affecting humans

Value to users

Wider **data**, **tools**, **secure environments** and **publication routes** for health research; cross-border and cross-domain studies **without moving the data**.

What the Node does

Intra-node: aligns the regional health data spaces and their secure processing environments (BIGAN, OHSIRIS, FEDERA Salud, HealthData@MAD-R&I) with the emerging Spanish health data space, the national EHDS node, and with the One Health data space at CESGA.

Inter-node: reaches the health data spaces of other Member States through HealthData@EU and the Nodes holding life-science data and distributed analysis.

How: discovery through (Health)DCAT-AP; analysis in secure processing environments or by script shipping (RAISE), with the CSIC privacy toolbox in the loop; data never leave the holder.

Builds on: the EHDS pilot of EOSC United, the propositions of the EOSC-A Health Data Task Force and EUCAIM.

The Node is not a Health Data Access Body and issues no permits.

Who

Lead: IACS

Partners: CSIC, BSC-CNS, CESGA, UPV, Biogipuzkoa, IBiS (HUVM, HUVR), IRYCIS, UA

Counterparts: Life Sciences Connect (EOSC ENTRUST), BBMRI-ERIC, EOSC EU Node

Plan in the build-up phase

- **M3** · implementation plan
- **M6** · mapping of FC-1/FC-2 to the Simpl stack
- **M15** · governed-access rehearsal on a permitted or synthetic cohort
- **M18** · end to end with counterparts; report ES-D5b

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ES-UC2 · Ocean Health

Long-term, multidisciplinary ocean observation turned into decision support

The challenge

The oceans regulate climate, the water cycle and atmospheric composition, and are **changing fast through warming, acidification, pollution and biodiversity loss**. The physical, chemical, biological and geological data needed to follow these changes are scattered across networks and formats.

Value to users

Patterns of ocean variability, **anticipation of ecosystem change and early warnings of extreme marine events**. The case is also the marine counterpart of the One Health scope of ES-UC1.

What the Node does

Platform: an intelligent, integrated and scalable platform that collects and analyses multidisciplinary ocean data and feeds operational decision-support systems.

Data: CSIC observation network with time series back to the 1930s, and NEREIDAS, the national marine-science data space demonstrator for the south-western Mediterranean (UMU), exposed as Node resources under one discovery and access path.

Analysis: AI methods applied to heterogeneous databases to optimise existing monitoring tools (AI4EOSC).

Interoperability: EMODnet, Copernicus Marine, Euro-Argo and EMSO ERICs, SeaDataNet.

Who

Lead: CSIC

Partner: UMU (NEREIDAS)

Counterparts: European Digital Twin of the Ocean Node; Blue-Cloud 2026 / CNR, already engaged with NEREIDAS

Plan in the build-up phase

- M3 · implementation plan and counterpart contact
- M9 · NEREIDAS and CSIC resources onboarded in the Node catalogue
- M18 · end to end with the counterpart Node; report ES-D5b

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ES-UC3 · Euclid DR1

A federated workspace for a collaboration whose membership changes every week

The challenge

A task force of the Euclid Consortium investigates **data analysis issues** while preparing Data Release 1. Its membership is dynamic and controlled by the task force itself — a pattern that federated infrastructure has to accommodate.

Value to users

A **distributed collaboration** obtains in days a federated shared workspace, with **reproducible and citable results**.

What the Node does

Storage: federated authentication and authorisation through eduGAIN and OIDC, elastic quota from 1 TB to 1,000 TB, access over HTTPS/WebDAV.

Analysis next to the data: JupyterHub for interactive work and CosmoHub for massive tabular exploration (200+ TB, 300+ billion objects).

Backbone: Rucio implementing the ESCAPE Data Lake architecture over RES storage, which shortens the integration path with the CERN Node.

Capabilities exercised: FC-1 with a real collaboration policy and FC-2 through IVOA-compliant catalogue exposure.

Who

Lead: PIC (Port d'Informació Científica)

Partner: CIEMAT

Counterpart: CERN Node, through the ESCAPE Data Lake and the shared Rucio backbone

Plan in the build-up phase

- M3 · implementation plan and counterpart contact
- M9 · federated workspace in operation for the task force (ES-MS8)
- M12 · end to end with report ES-D5a (ES-MS10)

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ES-UC4 · Parliamentary corpora

Comparable, multilingual parliamentary data for the social sciences and humanities

The challenge

Parliamentary debates are a primary source on **political discourse**. The proceedings of the Spanish Congress and the regional chambers, including those in the co-official languages, are not yet available in the comparable format used across Europe.

Value to users

Comparative study of **political discourse, terminological evolution** and the use of the co-official languages. The Spanish corpora join a family of comparable resources maintained by more than twenty countries, in a domain under-represented in the Federation.

What the Node does

Corpora: creation and enrichment of Spanish parliamentary corpora following the TEI schemes of ParlaMint, the CLARIN flagship for comparable parliamentary corpora.

Workflow: discovery in the Node catalogue and the CLARIN VLO; processing with the CLARIAH-ES annotation pipelines and open LLM inference for the official languages; deposit with PIDs in the CoreTrustSeal CLARIAH-ES repositories.

Capabilities exercised: FC-1, FC-2 and FC-7, and the composability sought by the EOSC Interoperability Framework: repository → tool → new research product.

Who

Lead: HiTZ (UPV/EHU), on behalf of CLARIAH-ES

Partners: IATEXT (ULPGC) and CLARIAH-ES centres

Counterparts: EOSC Node Netherlands (SURF), where ParlaMint is an EOSC use case with CLARIN ERIC; EOSC Node Slovenia (CLARIN.SI repository)

Plan in the build-up phase

- M3 · implementation plan and written contact with the counterpart Nodes
- M9 · CLARIAH-ES repositories and pipelines onboarded
- M12 · end to end with report ES-D5a (ES-MS10)

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The Node as users' gateway to the Federation

The concrete activities, who runs them and when

For researchers: one door

A researcher in Spain signs in with the institutional account and reaches resources of any Node; a researcher elsewhere finds Spanish resources through the **Federation catalogue**.

- Node catalogue harvested by the EOSC Resource Hub (FC-2) — RedIRIS, FECYT, UPM · build-up M6
- Single identity through RedIRIS SIR and eduGAIN (FC-1) · M6
- One helpdesk, with second-level support from resource owners and the CLARIN K-centre (FC-3) · UPV, IACS · M12

For providers: an open route in

Any Spanish institution can bring a resource to the Federation without running its own Federation interface.

- Published onboarding criteria and third-party contribution policy: TRL 7+, named owner, FAIR metadata · UPM, RedIRIS · M3
- Onboarding workshops on the metadata profile, validator and AAI integration · first wave M9
- Third-party onboarding opened at the first annual event · M12
- EOSC-A membership not required

For the community and policy makers

- Annual national EOSC-ES event, with the RES Users' Conference (over 1,000 researchers) as candidate co-location · M12, M24
- Node sessions in the public parts of EU project meetings hosted by partners in Spain, at no extra cost
- eosc.es domain reserved; landing page budgeted by IACS
- Structured dialogue with COS-Gob and the Ministry of Science, Innovation and Universities; national tripartite meeting on 2 October 2026

Evidence so far. Kick-off (May); first face-to-face meeting at CSIC Madrid on 29 June with about 22 institutions and speakers from NFDI, EOSC-CZ and the European DTO Node; online consolidation meeting on 23 July. Registrations for 2 October already include organisations outside the consortium: FECYT, CSUC, Consorcio Madroño, CREAF, Navarrabiomed, Fundación Ibercivis and Universidad de Zaragoza.

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Stakeholder engagement and KPIs

Different messages and channels per profile, measured against the Project Plan (D1)

Stakeholder	What they gain from the Node	Main channel
Researchers and research teams	One route to Spanish and European data, services and compute	Institutions' channels, annual event, helpdesk, training
Research organisations, universities, libraries and repositories	European visibility without operating their own Federation interface	Onboarding route and workshops, RECOLECTA network
Service and infrastructure providers	A wider user base through shared Federating Capabilities	Technical committee, onboarding framework
Public bodies and funders	An instrument to implement and monitor ENCA and PEICTI	Strategic committee, COS-Gob, tripartite
Industry and citizens	Results of data spaces addressing health, environmental and societal challenges	Data spaces and use case communications

KPIs of the Project Plan (D1)

KPI 6 · 2 face-to-face and 2 online meetings: 29 June F2F and 23 July online held; 2 October F2F with the tripartite; final online meeting in October.

KPI 7 · 10 co-applicants in EOSC Academy content: 11 institutions followed EOSC Academy courses — met.

KPI 9 · 3 alignment actions with data spaces: use cases mapped to four Common European Data Spaces and a Simpl alignment plan in the Charter; to be reported in November.

Tracking indicators

- Meetings held, in person and online
- Institutions reached, EOSC-A members and non-members: 22 founding institutions, 8 of them outside EOSC-A
- Meetings with policy makers: COS-Gob (June), tripartite with the Ministry (October)

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Training and the EOSC Academy

What the consortium learned, what the Node will teach, what it gives to the EOSC Academy

Done: capacity built in the consortium

11 institutions followed EOSC Academy courses (D1 **target: 10**):

- Course 1 · Explaining the EOSC Federation — Biogipuzkoa, UJA
- Course 2 · Applying the EOSC Federation Handbook — BSC, Biogipuzkoa, CESGA, CSIC, IACS, IMIB, RedIRIS, UMU
- Course 3 · Use cases for the EOSC Federation — Biogipuzkoa, CSIC, EHU, IBiS-HUVR, UMU
- Course 7 · The EOSC AAI explained — Biogipuzkoa, UMU

Build-up: the Node's training offer

- Spanish CLARIN K-centre (CLARIAH-ES): tutorials, course materials and expert advice
- Onboarding workshops for providers: metadata profile and validator (FC-2), integration with the EOSC Federated AAI (FC-1), monitoring probes and FitSM procedures
- First training session to the Node community at build-up M9 (ES-MS7)
- Engagement and training report (ES-D6a, M12); peer-learning workshops and infrastructure sandboxes (ES-D6b, M24)

Contribution to the EOSC Academy

November: the EOSC Academy matrix lists the training activities of the consortium with format, audience, reusability, infrastructure, maturity and readiness for the EOSC Academy.

Skills gaps already visible during preparation, to be recorded in the matrix: AAI integration; metadata profiles and harvesting for Federation catalogues; helpdesk integration across providers; FitSM service management; governed access to sensitive data in secure processing environments.

Build-up: reusable materials, sandboxes and workshop content packaged for the EOSC Academy (ES-D6b, M24).

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Lessons learned

From other Nodes and from our own process — and how each lesson changed the plan

Each Node must fit its national reality	**Learned:** NFDI is one association of 327 members; EOSC-CZ is led by a single e-infrastructure. Spain has no such body, and 22 institutions are building the Node bottom-up.	**Changed:** An existing entity signs; an open collaboration agreement gives every institution a voice, so the Node does not “belong” to one organisation.
Build on what already operates	**Learned:** Nodes that grew out of operating communities (European DTO / Blue-Cloud) reached production faster than those assembled administratively.	**Changed:** Every mandatory capability is mapped onto a running national service; use cases are anchored in communities that already collaborate.
Start small, then grow	**Learned:** Nodes are best built from a few services integrated first; onboarding everything at once fails.	**Changed:** 12 resources and 4 use cases selected from 28 resources and 10 candidate cases; the rest stay in a public pipeline.
Bottom-up must meet top-down	**Learned:** Enrolment requires endorsement by the national authority; regional and national priorities must be aligned.	**Changed:** Early briefing of COS-Gob and a national tripartite meeting with the Ministry before the enrolment application.
Volunteer effort starts a Node but does not run it	**Learned:** Coordination overhead and turnover were the risks most cited on 29 June.	**Changed:** Named roles with declared effort; “no named owner, no onboarding”; decision records.

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Sustainability and the way forward

Business model of the future Node and steps to the enrolment application

Business model

Operation in kind. Each institution bears the cost of the resources it contributes to Node Exchange; the Node adds a thin federating layer, not new infrastructure.

Core services on existing budgets. Identity (RedIRIS), catalogue harvesting (FECYT), compute (RES) keep their owners and funding lines.

Shared costs. The collaboration agreement sets membership, cost-sharing and exit rules; funders sit in the strategic committee.

Build-up funding. COFUND HORIZON-INFRA-2027-01-EOSC-01 through a national funding agency, national calls under PEICTI, and partners' Horizon Europe projects.

Long term. EOSC-ES in the national science, technology and innovation plans for 2028–2034.

Next steps

2 Oct 2026	Tripartite meeting, Zaragoza: governance model, candidates for representing entity, role holders, core-service providers, use case selection
Oct 2026	Final online meeting of the preparatory project: readiness assessment and next steps
Early Nov 2026	Use case report, EOSC Academy matrix and showcase presentation
Before enrolment	Collaboration agreement signed; representing entity and five role holders named; endorsement by the national authority (ES-MS0)
Early 2027	Application to the third enrolment wave (expected)
Build-up M1–M24	Node Core in production at M6, first onboarding wave at M9, use cases demonstrated at M12 and M18

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